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Alleviation of propionic acid inhibition and enhancement of methane production in anaerobic digestion by micronano-bubble water supplementation

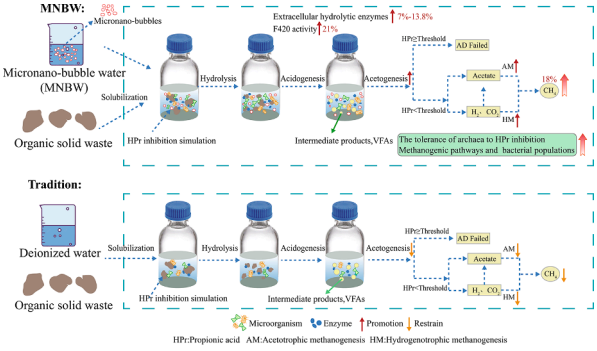
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HIGHLIGHTS

- Micronano-bubble water (MNBW) addition accelerated propionic acid (HPr) degradation.
- MNBW boosted partial and total alkalinity in anaerobic digestion system.
- Methane production was enhanced by 13.5%–18.4% with MNBW under HPr inhibition.
- MNBW supplementation promote hydrolysis and methanogenesis under HPr inhibition.
- MNBW improved the tolerance of archaea such as *Methanosaetaceae* to HPr inhibition.

GRAPHICAL ABSTRACT



ARTICLE INFO

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ABSTRACT

Micronano-bubble water (MNBW) was used to accelerate propionic acid (HPr) degradation and enhance methane production in anaerobic digestion (AD) in this study. Results showed that HPr degradation was accelerated by MNBW addition. Moreover, methane production was elevated by 13.5% to 18.4% under HPr inhibition by MNBW supplementation. Microbial analysis revealed that MNBW boosted the bacterial populations involved in protein and carbohydrate consumption. In addition, supplementation with MNBW improved the tolerance of archaea such as *Methanosaetaceae* to HPr inhibition, thus increasing methane production. MNBW also promoted alkalinity recovery, lowered intermediate alkalinity/partial alkalinity and volatile fatty acid/total alkalinity ratios, and enhanced buffering capacity. Enzyme activity increased by MNBW, including a 21.0% increase in coenzyme F420 and a 7.0% to 13.8% increase in extracellular hydrolase activity. This work highlights that MNBW is a promising approach to accelerate HPr degradation, optimize methanogenic pathways, and boost methane production.

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1. Introduction

Among the various biomass waste treatment technologies, anaerobic digestion (AD) is widely recognised for its ability to reduce pollution and produce bioenergy, making it applicable across multiple industries (Mo et al., 2023). AD offers significant advantages including renewable energy generation, pathogen inactivation, and odour control (Xu et al., 2023). However, its effectiveness is often limited by inhibitory factors, with volatile fatty acids (VFAs) accumulation being one of the main causes of decreased efficiency and lower energy yield. VFAs inhibition is a common issue in dry AD process, particularly during the acidogenesis stages, when the overproduction of organic acids leads to the accumulation of VFAs (Li et al., 2024b).

As one of the main components of VFAs, propionic acid (HPr) is converted to acetate before being consumed by syntrophic propionate oxidizing bacteria in the hydrogenotrophic pathway (Eq. (1) to (3) (Rocamora et al., 2023). The process of HPr oxidation to acetate is prone to inhibition problems, because this process is the most thermodynamically unfavoured reaction in the chain (Stams and Plugge, 2009).

$$CH_3CH_2COO^- + 2H_2O \leftrightarrow CH_3COO^- + CO_2 + 3H_2 \quad (1)$$

$$CH_3COO^- + H^+ + 2H_2O \leftrightarrow 2CO_2 + 4H_2 \quad (2)$$

$$4H_2 + CO_2 \leftrightarrow CH_4 + 2H_2O \quad (3)$$

HPr is particularly challenging to degrade because of its high energy demand, which makes it thermodynamically unfavourable (Cruz Viggi et al., 2014). According to a previous study, the accumulation of HPr is due to thermodynamic blocking of HPr degradation caused by hydrogen partial pressure exceeding 10^{-4} bar (Rocamora et al., 2023). However, some reports suggest that the increased hydrogen partial pressure is not the cause of HPr accumulation, but rather a by-product of HPr degradation (Choong et al., 2016; Banks et al., 2012). A lack of trace elements like Fe, Ni, Se, W or Co is another cause of HPr accumulation, as they enable HPr degradation and the hydrogenotrophic pathway, and are necessary for enzymes and direct interspecies electron transfer (DIET) required for the relationship between methanogens and syntrophic bacteria (Westerholm et al., 2015; Li et al., 2021a). The abundance of methanogens consuming acetic acid is influenced by the conversion rate of HPr and is ultimately reflected in biogas production (Han et al., 2020). Accumulation of HPr can lead to acidification of AD system, decrease of methanogens activity, and eventually lead to AD system collapse (Li et al., 2021b). Therefore, effective HPr degradation is critical for overcoming HPr inhibition and improving AD efficiency. Various strategies, such as the use of nano-magnetite (Li et al., 2021b), nanoscale zero-valent iron (Cheng et al., 2020), and $FeCl_3$ (Yang et al., 2024), have been explored to enhance HPr breakdown. Although promising, these approaches are often expensive, environmentally unsustainable, and difficult to scale for industrial applications. It is urgent to develop an alternative approach that can fundamentally accelerate HPr degradation with a low cost, no secondary pollution, and convenient operation.

One promising technology is micronano-bubble water (MNBW), which contains high concentrations of nanobubbles (Liu et al., 2019). These micronano-bubbles (MNBs) are gas-filled cavities with diameters range from a few nanometers to hundreds of nanometers that exhibit unique characteristics (Ghaani et al., 2020). MNBW is widely used in many fields, such as medicine and agriculture (Bismuth et al., 2022; Marcelino et al., 2023). Previous report has suggested that MNBW is beneficial for increasing methane yield (Yang et al., 2019). Fan et al. (2020) applied three kinds of MNBW (Air, H_2 and CO_2) to the AD of SM and obtained preliminary experimental results that all three kinds of MNBW can significantly increase methane production. Among them, Air-MNBW has the lowest cost and the simplest operation. Hou et al. (2021) found that the promotion effect of Air-MNBW on methane production was higher than that of N_2-MNBW. Therefore, Air-MNBW can facilitate the AD process while being cost-effective and easy to operate. Although the conclusion that MNBW can promote AD has been confirmed, these preliminary findings were based on the premise that MNBW does not appear to inhibit AD. Given the unique properties of MNBW and its positive performance in AD systems, MNBW may have the potential to alleviate HPr inhibition. In addition, the use of MNBW to accelerate HPr degradation without secondary pollution and the preparation cost of MNBW are low, easy to operate, and has the potential for industrial application.

Therefore, this work investigated the use of MNBW to accelerate HPr degradation, alleviate VFAs inhibition, and improve methane production. This study represents the first exploration of the application of MNBW to HPr degradation under inhibitory conditions and provides a potential solution to one of the key challenges in AD technology. By addressing HPr accumulation, this study aims to contribute significantly to the field of renewable energy production through more efficient AD processes.

2. Materials and methods

2.1. Preparation of substrate, inoculum and MNBW

2.1.1. Substrate and inoculum preparation

In this study, to effectively treat organic waste and prevent environmental pollution, we selected swine manure (SM) as the substrate because the substrate used in AD should be readily available and cost-effective. The SM was obtained from Guigang pig farm, Guangxi Province, China. To avoid interference from foreign materials during the experiments, impurities such as pig feed were removed from the SM. Additionally, to ensure consistency across different experimental groups, the SM samples were thoroughly homogenised before use in subsequent AD experiments. The digested sludge samples used as inoculum were sourced from Guangzhou sewage treatment plant, Guangdong Province, China. The acclimation procedure was as follows: SM collected from the same farm and digested sludge were mixed at a feed-to-microorganism ratio of 1:1 (volatile solid [VS] basis) and placed in 1-L Schott Duran bottles. The characteristics of the AD substrate and inoculum are listed in Table 1.

2.1.2 Preparation of MNBW

MNBW was prepared using a 6-L MNBW generator (Yuchuang Environmental Technology Co., Ltd., Shandong, China), which was described in detail in the previous study (Tang et al., 2022). Considering the cost and practical application of organic waste treatment, air was selected for the preparation of MNBW (Air-MNBW), allowing us to source air directly from the natural environment, without the need for more expensive gases such as oxygen, nitrogen, or hydrogen. Air was introduced into the MNBW generator following a procedure described previously (Tang et al., 2022; Yang et al., 2019). The flow rates of air and deionized water (DW) were set to 0.6 L/h and 3 L/min respectively. The pressure was maintained at 0.24 ± 0.01 MPa. The temperature was set at 20 ± 1°C, and the preparation time was 30 min. It can be observed that MNBW changes from a milky opaque liquid to a colorless transparent liquid. The prepared MNBW was subjected to a subsequent AD process involving SM.

2.2. HPr inhibition batch experimental set-up

In this study, the batch experiments utilized 250 mL Schott Duran bottles. The raw substrate and acclimated inoculum were loaded into 250 mL reactors (R) with a working volume of 200 mL, maintaining a feed-to-microorganism ratio (on a VS basis) of 1:1. Each reactor contained 4.65 g SM and 95.35 g inoculum, with a total loading of 100 g. Subsequently, 100 mL MNBW was supplemented to MNBW groups and the same volume of DW was supplemented to control groups. To

simulate varying degrees of HPr inhibition, equal concentrations (1000 mg/L) of the HPr standard solution were added to the reactors at different volumes according to the results of the preliminary experiment. In addition, in order to better set the HPr inhibition gradient of this work, some previous research reports were referred to (Yang et al., 2024; Li et al., 2021). HPr were added to achieve final concentrations of 1.25, 2.5, and 5.0 g-HPr/L. According to the final concentration of HPr obtained and whether MNBW was added, the batch reactors were grouped into eight groups, named Control-0, Control-1.25, Control-2.5, Control-5, MNBW-0, MNBW-1.25, MNBW-2.5, and MNBW-5. In addition, two control bottles containing only the inoculum were tested for comparison with the four experimental groups (see supplementary material).

To simulate different levels of HPr inhibition and avoid interference from acidic or alkaline chemicals, no pH adjustments were made during the experiment. This also prevented potential contamination by chemical additives. After ensuring thorough mixing of the materials within the reactors, nitrogen (99.9995 %) was applied to purge the 250 mL bottles for 3 min. The reactors were sealed using specialised caps. Before placing all the reactors in an incubator (35 ± 2 °C), all reactors were checked to ensure they were properly sealed, and the batch AD experiment was initiated. Three parallel and two replicate experiments were conducted. The key AD parameters were measured at regular time points.

2.3. Methods of detection and analysis

In this experiment, biogas production was obtained using 50 and 5 mL syringes and normalized to standard temperature and pressure (1 atm, 25 °C). The biogas composition, including methane and other gas components, was analysed according to a previous study with a sample volume of 1 mL (Yang et al., 2019). The total solid, VS and total ammonia nitrogen (TAN) were determined based on the standard methods (APHA 2012). Soluble proteins and carbohydrates were measured according to the methods reported in the previous literature (Fan et al., 2020). The concentrations of HPr and other components of VFAs were analysed according to a previous report (Yang et al., 2019). Total alkalinity (TA), intermediate alkalinity (IA), and partial alkalinity (PA) were determined according to previously reported methods (Banks et al., 2012).

To assess the enzymatic activity during the hydrolysis phase of AD, the activities of acid phosphatase, α-glucosidase, alkaline phosphatase, and protease were determined on day 4, following the methodologies outlined in an earlier report (Goel et al., 1998). Additionally, F420, which is closely associated with methanogenic microorganisms, was determined according to previously established methods (Gao et al., 2024). At the end of the experiment, AD sludge samples under different HPr inhibition were collected and the microbial community structure was subsequently analyzed. During microbial analysis, the E.Z.N.A. Soil DNA Kit (OMEGA) was applied to extract genomic DNA. The NanoDrop 2000 spectrophotometre (NanoDrop Technologies) was used to determine the purity and concentration (Tang et al., 2024). Samples were collected from the reactors for high-throughput sequencing. The different sets of primers were used to amplify 16S rRNA for the analysis of archaeal communities (524F10extF (5′-TGYCAGCCGCCGCGGTAA-3′)/Arch958RmodR (5′YCCGGCGTTGAVTCCAATT-3′)) and bacterial communities (338F('5-ACTCCTACGGGAGGCAGCAG-3′)/806R (5′-GGACTACHVGGGTWTCTAAT-3′)). In addition, the raw sequencing data has been uploaded to public database. The SubmissionID is SUB15017517 and the BioProject ID is PRJNA1212302.

Table 1
Characteristics of raw swine manure and inoculum used in this study.

Parameters	Unit	Inoculum Raw swine manure
Total solids	%	1.47 ± 0.06 26.77 ± 0.64
Volatile solids	%	1.09 ± 0.05 22.37 ± 0.39
Soluble total carbon	mg/g	ND 36.59 ± 0.89
Soluble inorganic carbon	mg/g	ND 4.17 ± 0.21
Soluble total organic carbon	mg/g	ND 32.43 ± 0.96
Volatile fatty acids	mg/g	ND 32.15 ± 0.78
Total alkalinity (as $CaCO_3$)	mg/g	1.92 ± 0.12 14.80 ± 0.36
Soluble proteins	mg/g	0.31 ± 0.02 11.79 ± 0.53
Soluble carbohydrates	mg/g	0.11 ± 0.01 5.56 ± 0.30
Total ammonia nitrogen	mg/g	0.51 ± 0.03 3.96 ± 0.42
pH	–	7.38 ± 0.06 6.39 ± 0.04

ND (Not Determined).
Data in the table are expressed as Means ± SD.

2.4. Kinetic models analysis

In this study, the modified Gompertz model (Eq. (4) was applied to perform kinetic analysis of biogas and methane production process based on the previous report (Karki et al., 2022).

$$P = P_{max} exp\left\{ -exp\left[\frac{R_{max}e}{P_{max}}(\lambda - t) + 1\right]\right\} \quad (4)$$

In Eq. (4), P is the cumulative biogas or methane production (mL/g-VS), P_{max} is the maximum biogas or methane production potential (mL/g-VS), R_{max} is the maximum biogas or methane production rate (mL/g-VS·d), λ is the lag phase (d), t is the duration of the AD process (d).

2.5. Statistical analysis

In this study, experimental data on the MNBW promotion of HPr degradation and mitigation of acid inhibition were analysed using Microsoft Office Excel 2021. The mean ± standard deviation (SD) was applied to facilitate subsequent discussion. One-way analysis of variance followed by Tukey's post-hoc test was performed to assess statistical differences. Graphing and further data analyses were conducted using Origin 9.0.

3. Results and discussion

3.1. Variations in HPr and VFAs concentrations

HPr is the main component of VFAs, and its trend during AD is closely related to VFAs concentration. Overall, except MNBW-5 and Control-5, the VFAs concentration of the other reactors increased first and then decreased (Fig. 1a and 1b). For MNBW-5 and Control-5, the VFAs concentration remained above 5000 mg/L, indicating severe VFAs inhibition. The VFAs concentration of MNBW-2.5 was higher than Contol-2.5 on day 4 ($p < 0.05$), but significantly lower on days 16 and 22. A similar pattern was observed in Control-1.25/MNBW-1.25 and Control-0/MNBW-0. These results suggest that MNBW facilitates hydrolysis, thereby accelerating VFAs production. As the AD process continued, MNBW accelerated the consumption of VFAs, thereby preventing their accumulation.

The degradation of HPr is thermodynamically considered an unfavourable reaction, and its accumulation is the primary cause of VFAs inhibition (Rocamora et al., 2023). To investigate the impact of MNBW on HPr degradation, the HPr concentration was measured across the different treatment groups, as illustrated in Fig. 1a and 1b. No statistically significant differences were observed between the $R_{Control}$ and R_{MNBW} in the same HPr treatment group at the beginning of the AD process ($p > 0.05$). The HPr concentrations in Control-0/MNBW-0 initially increased before decreasing, with HPr concentrations remaining below 700 mg/L, indicating minimal HPr accumulation. Notably, MNBW-0 was significantly lower than Control-0 by 24.2 % on day 4 ($p = 0.004 < 0.05$), and 58.2 % on day 16 ($p < 0.001$). Furthermore, MNBW-1.25 exhibited significantly lower HPr concentrations than that in Control-1.25 on days 10 and 16. Specifically, MNBW-1.25 was significantly lower than MNBW-1.25 by 21.0 % on day 10 ($p =$

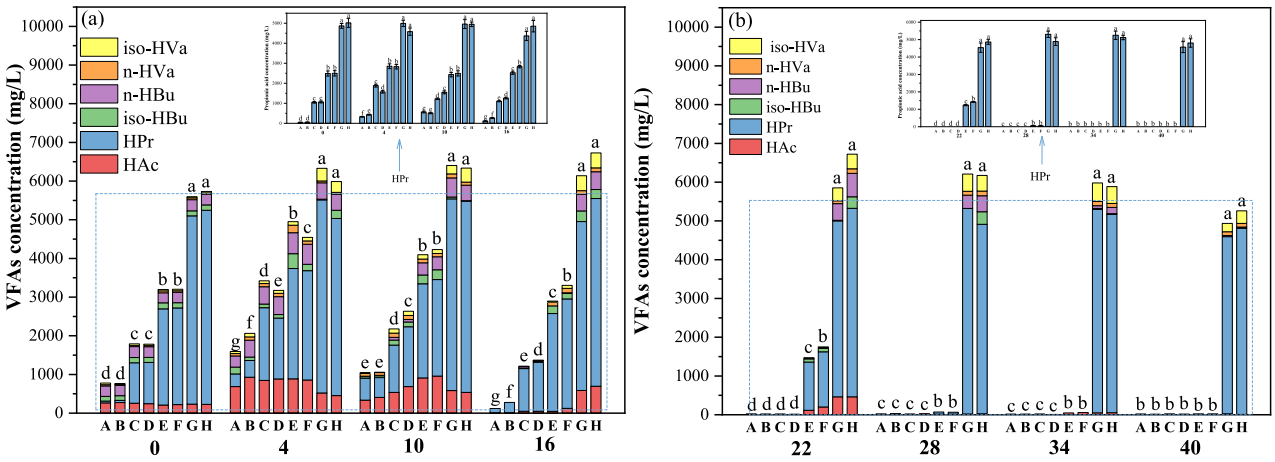


Fig. 1. Changes in VFAs and propionic acid concentrations in AD process. The alphabets under X-axis areas: A(MNBW-0), B(Control-0), C(MNBW-1.25), D(Control-1.25), E(MNBW-2.5), F(Contol-2.5), G(MNBW-5), H(Control-5). The different letters (a, b, c, …) indicate significant difference at the 5 % level ($p < 0.05$).

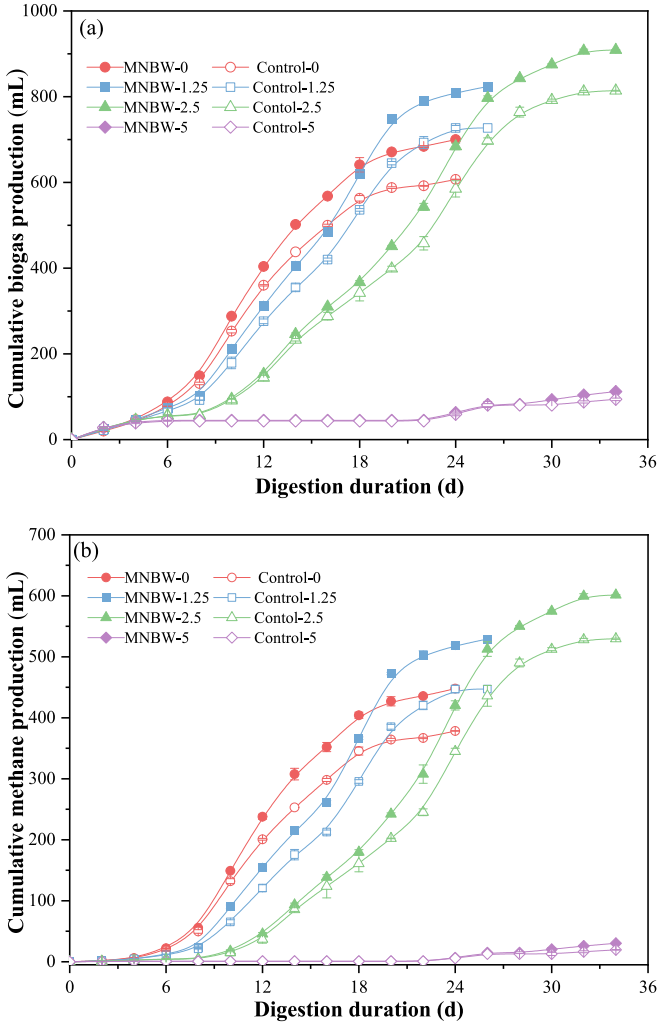


Fig. 2. (a) Cumulative biogas production. (b) Cumulative methane production.

0.002 < 0.05), and 12.5 % on day 16 ($p = 0.004 < 0.05$). MNBW-2.5 also demonstrated significantly lower HPr concentrations than that in Contol-2.5 on days 16 and 22 ($p < 0.05$). Among them, MNBW-1.25 was significantly lower than MNBW-1.25 by 10.7 % on day 16 ($p = 0.006 <$ 0.05), and 12.6 % on day 22 ($p = 0.002 < 0.05$). Moreover, a pronounced enhancement of HPr degradation by MNBW-2.5 was observed on days 16 and 22, which occurred later than the effects seen with MNBW-1.25 and MNBW-0, indicating that as the concentration of HPr increased, the promoting effect of MNBW on HPr degradation diminished. Additionally, Control-5/MNBW-5 maintained HPr concentrations exceeding 4300 mg/L throughout the AD process, with no statistically significant differences between MNBW-5 and Control-5. This result showed that when the HPr concentration exceeded the threshold, MNBW failed to effectively accelerate HPr degradation. In summary, at low HPr concentrations (Initial HPr = 1.25 and 2.50 mg/L), MNBW significantly enhanced HPr degradation. However, as the HPr concentration increased, this promoting effect gradually weakened. Once the HPr concentration surpassed the threshold, the supplementation of MNBW did not lead to effective degradation of HPr.

3.2. *Biogas and methane production*

The biogas production data obtained under varying degrees of HPr inhibition are shown in Fig. 2a. The maximum biogas production occurred in MNBW-2.5, reaching 909 ± 22 mL. This result showed an 11.7 % increase relative to the Contol-2.5. The biogas yield of MNBW-1.25 was 824 ± 19 mL, reflecting a 13.3 % increase over that of the Control-1.25. In the absence of HPr inhibition, MNBW also enhanced biogas production, with MNBW-0 yielding 700 ± 15 mL, which is a 15.3 % increase over that of the Control-0. The increase in biogas production attributable to MNBW across different HPr concentration treatment groups suggests that the promoting effect of MNBW diminished as the severity of HPr inhibition increased. Notably, biogas production in the Control-5/MNBW-5 groups was significantly reduced compared to the other groups, with no significant difference observed between MNBW-5 and Control-5, indicating that MNBW does not significantly enhance biogas yield when the HPr concentration exceeded the threshold. The kinetic analysis results of biogas production process are shown in Table 2. The maximum biogas production potential and the maximum biogas production rate of MNBW-0, MNBW-1.25, and MNBW-2.5 were higher than those of the corresponding control group. Moreover, the lag phase of MNBW-0, MNBW-1.25, and MNBW-2.5 was shorter than that of the corresponding control group.

The trend of methane accumulation during the AD process mirrors that of biogas production (Fig. 2b). The cumulative methane yield of MNBW-0, MNBW-1.25, and MNBW-2.5 was 448 ± 8, 529 ± 10, and 601 ± 14 mL, respectively, indicating increases of 18.4 %, 18.3 %, and 13.5 %, compared with Control-0, Control-1.25, and Contol-2.5, respectively. These results confirmed the promoting effect of MNBW on cumulative methane yield. The kinetic analysis results of methane production

Table 2
Kinetic analysis of cumulative biogas production with and without micronano-bubble water supplementation.

Models	Control-0	MNBW-0	Control-1.25	MNBW-1.25	Control-2.5	MNBW-2.5
Modified Gompertz	0.9988	0.9960	0.9939	0.9932	0.9899	0.9916
R^2	5.23 ± 0.21^{c}	4.65 ± 0.17^{d}	6.30 ± 0.40^{b}	6.35 ± 0.43^{b}	8.77 ± 0.56^{a}	9.24 ± 0.52^{a}
λ (d)	329.37 ± 13.61^{e}	362.32 ± 9.05^{d}	429.88 ± 23.98^{c}	473.18 ± 25.50^{bc}	532.01 ± 45.26^{ab}	557.88 ± 36.77^{a}
Pmax (mL/g-VS)	23.65 ± 1.45^{abc}	25.53 ± 1.01^{a}	22.63 ± 1.17^{bc}	26.37 ± 1.51^{a}	18.55 ± 0.91^{d}	21.94 ± 1.12^{c}
Rmax (mL/g-VS·d)						

Data in the table are expressed as Means ± SD, and the different letters (a, b, c, …) indicate significant difference at the 5% level ($p < 0.05$).

Table 3
Kinetic analysis of cumulative methane production with and without micronano-bubble water supplementation.

Models	Control-0	MNBW-0	Control-1.25	MNBW-1.25	Control-2.5	MNBW-2.5
Modified Gompertz	0.9997	0.9977	0.9969	0.9975	0.9939	0.9988
R^2	6.36 ± 0.20^{c}	6.57 ± 0.13^{c}	8.74 ± 0.28^{b}	8.52 ± 0.20^{b}	12.30 ± 0.43^{a}	12.56 ± 0.42^{a}
λ (d)	189.21 ± 1.58^{e}	220.28 ± 4.15^{d}	251.40 ± 12.21^{c}	278.98 ± 9.00^{b}	306.63 ± 5.03^{a}	319.86 ± 9.41^{a}
Pmax (mL/g-VS)	16.85 ± 0.55^{bc}	22.38 ± 1.09^{a}	15.56 ± 0.77^{c}	21.15 ± 1.10^{a}	13.01 ± 0.41^{d}	18.39 ± 1.22^{b}
Rmax (mL/g-VS·d)						

Data in the table are expressed as Means ± SD, and the different letters (a, b, c, …) indicate significant difference at the 5% level ($p < 0.05$).

process are shown in Table 3. The maximum methane production rate of MNBW-0, MNBW-1.25, and MNBW-2.5 was 22.38 ± 1.09, 21.15 ± 1.10, and 18.39 ± 1.22 mL/g-VS·d, respectively, indicating increases of 32.8 %, 35.9 %, and 41.4 %, compared with Control-0, Control-1.25, and Contol-2.5, respectively. In addition, the maximum methane production potential of MNBW-0, MNBW-1.25, and MNBW-2.5 was also higher than that of the corresponding control group. Previous report has also indicated that MNBW can enhance the methane yield (Fan et al., 2020). Compared to this previous report, the lower methane increment in this work was primarily because of the cost-saving decision to use Air-MNBW instead of H_2-MNBW or other kinds of MNBW. The superior performance of H_2-MNBW in promoting methane production may stem from its utilisation as a substrate by methanogens, thereby further increasing the methane yield.

Given identical substrate (raw SM) and inoculum loading, MNBW-2.5 achieved the highest biogas and methane yield, owing to the degradation and utilisation of HPr added to simulate inhibition at the beginning of the AD process. For Control-5/MNBW-5, the minimal degradation of HPr resulted in VFAs accumulation (VFAs concentrations exceeding 5000 mg/L), significantly affecting methane production. Furthermore, no statistically significant difference was observed between MNBW-5 and Control-5 in the methane yield ($p > 0.05$).

Subsequently, economic evaluation was applied in this work to better evaluate the economic potential of MNBW (see supplementary material). MNBW-0, MNBW-1.25 and MNBW-2.5 obtained 536.04, 633.05 and 719.61 kWh of electric energy by processing 1 ton of SM, respectively, which were significantly higher than their corresponding control group ($p < 0.05$). These results indicate that NBW supplementation can significantly improve the economy of AD of SM under HPr inhibition. In addition, the electrical energy required for preparing MNBW to treat 1 ton of SM was calculated in this research. The MNBW generator used in this work uses the latest energy efficient design in

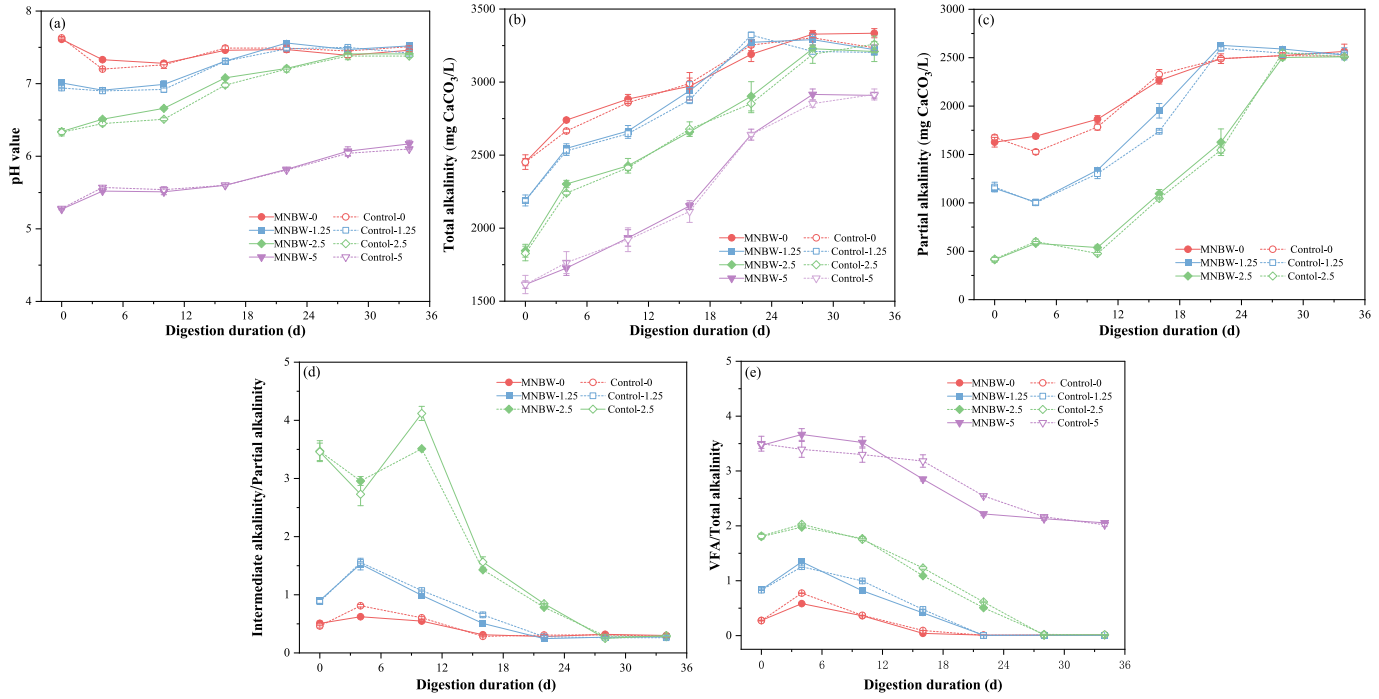


Fig. 3. Performance of the buffering capacity. (a) pH value. (b) Total alkalinity. (c) Partial alkalinity. (d) Intermediate alkalinity/Partial alkalinity. (e) VFA/Total alkalinity.

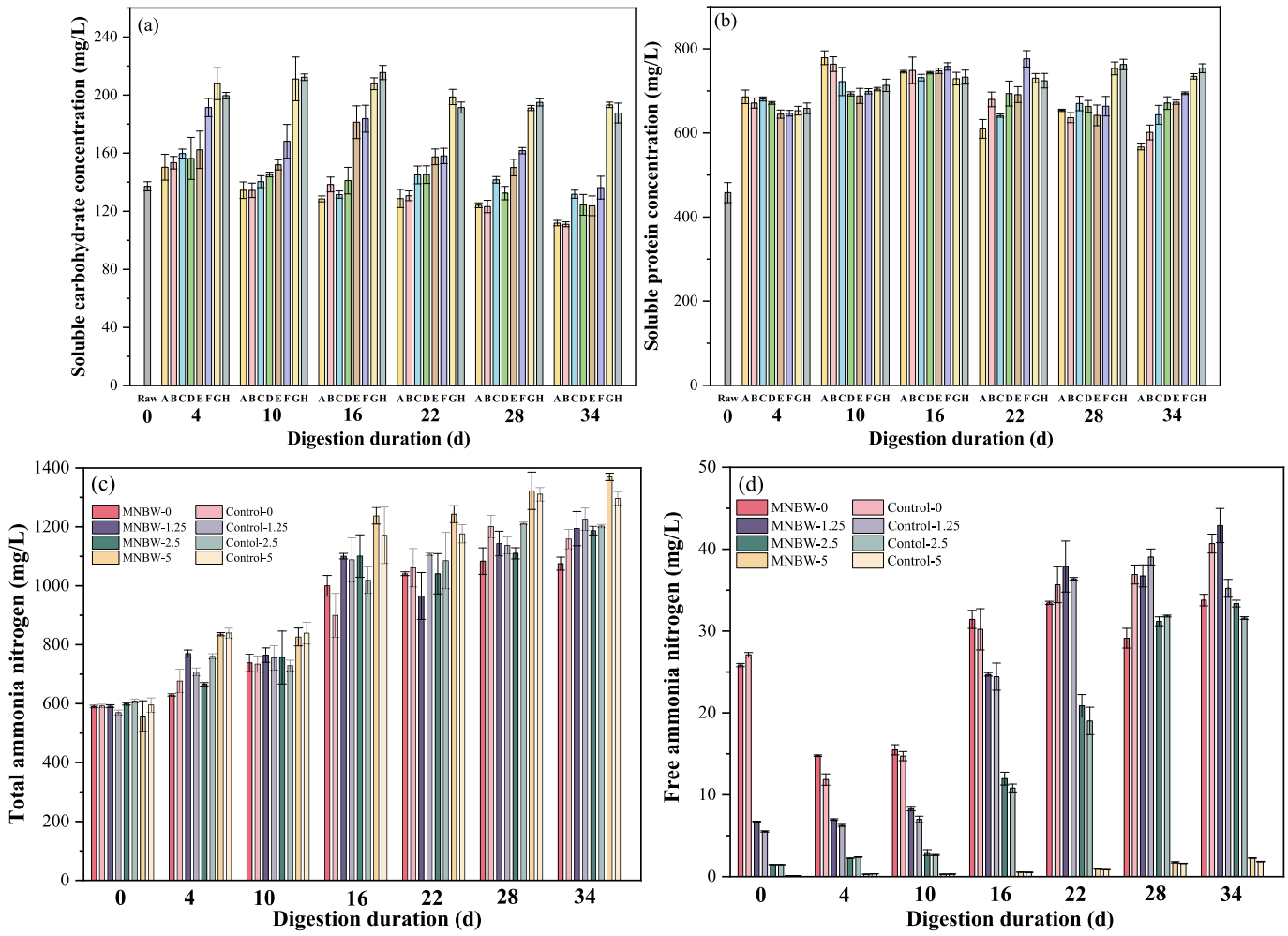


Fig. 4. Soluble organics and TAN manifestations. (a) Soluble carbohydrate concentration. (b) Soluble protein concentration. (c) Total ammonia nitrogen. (d) Free ammonia nitrogen. The alphabets under X-axis areas: A(MNBW-0), B(Control-0), C(MNBW-1.25), D(Control-1.25), E(MNBW-2.5), F(Contol-2.5), G(MNBW-5), H (Control-5).

2024 and has a power of 0.072 kW, capable of producing 150 L MNBW in one hour. The power consumption of MNBW generator in the process of treating 1 ton of SM was 10.33 kWh after calculation. Although the production of MNBW also consumes electrical energy, its consumption was much lower than its output. Therefore, under HPr inhibition, the approach of alleviating HPr inhibition and increasing methane production by supplementing MNBW has economic value and is worthy of further study in the future.

In summary, the gas production performance of the reactors under different degrees of HPr inhibition indicated that as the level of HPr inhibition increased, the promoting effects of MNBW gradually diminished. In addition, the supplementation of MNBW significantly increased the methane production in AD of SM under HPr inhibition, which has economic value. MNBW failed to significantly enhance gas production when HPr concentrations exceeded the inhibition threshold.

3.3. Performance of the buffering capacity

The pH value is an important parameter in AD systems and serves as an indicator of system stability (Wu et al., 2022). The pH value of all bottles was tested throughout the AD process. The initial pH values differ among the reactor groups owing to the varying concentrations of HPr (Fig. 3a). Control-0/MNBW-0 and Control-1.25/MNBW-1.25 exhibited an initial decline, followed by gradual increases, ultimately stabilising. The initial decline in pH was mainly due to the production of VFAs, whereas the subsequent increase in pH was due to the consumption of VFAs. The pH of Contol-2.5/MNBW-2.5 increased gradually and stabilised, indicating that the AD process experienced initial HPr inhibition, which was subsequently alleviated as VFAs were gradually consumed. Moreover, the pH of MNBW-2.5 on days 4, 10, and 16 were significantly higher than Contol-2.5. In contrast, MNBW-5 and Control-5 were below 5.3 at the beginning of the AD process, and the pH remained below 6.2 throughout the AD process, suggesting HPr inhibition not being fully alleviated during the 34 days.

Alkalinity is a key indicator of the stability of AD systems and numerous studies have established its relationship with system stability (Zheng et al., 2021; Shen et al., 2023). Fig. 3b illustrates the changes in the total alkalinity (TA) content across all reactors. Control-5/MNBW-5 exhibited the lowest TA content, approximately 1614 ± 63 mg $CaCO_3$/L at the onset of the AD experiment, indicating a relatively poor buffering capacity. Conversely, Control-0/MNBW-0 had the highest TA content, followed by Control-1.25/MNBW-1.25 and Contol-2.5/MNBW-2.5. The TA levels increased in all groups, with Control-5/MNBW-5 consistently exhibiting significantly lower TA levels than the other groups, indicating poor stability. Notably, Control-0 were lower than MNBW-0 on days 4 and 10, suggesting that MNBW contributed to increasing the TA levels. Similar patterns were observed for MNBW-1.25 and MNBW-2.5, which demonstrated higher TA levels than Control-1.25 and Contol-2.5 on days 4, 10, 16 and on days 4, 10, respectively. Specifically, the TA content of MNBW-0, MNBW-1.25 and MNBW-2.5 on day 4 was 2740 ± 53, 2546 ±

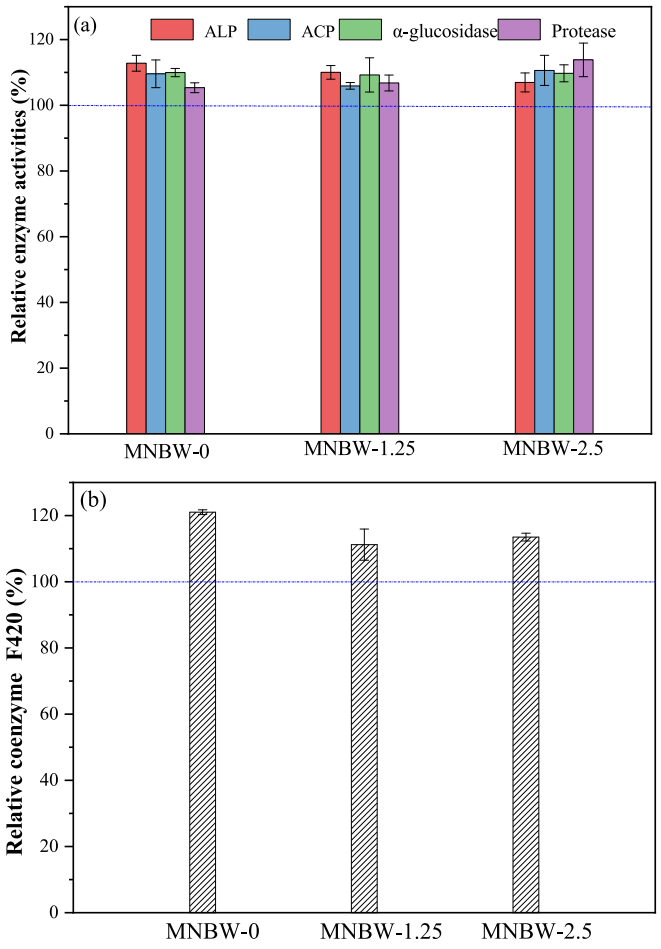


Fig. 5. Enzyme activity manifestations. (a) Relative activities of four types of extracellular hydrolytic enzymes. (b) Relative activities of coenzyme F420.

41, and 2302 ± 44 mg $CaCO_3$/L, respectively, indicating increases of 4.8 %, 2.3 % and 4.7 %, compared with Control-0, Control-1.25, and Contol-2.5, respectively. These results indicate that, despite HPr inhibition, MNBW aids in increasing TA levels, thus improving the buffering capacity.

The partial alkalinity (PA) parameter is also a significant indicator of AD system stability (Wu et al., 2021). The PA concentrations gradually increased and stabilised, mirroring the trend observed for TA (Fig. 3c). MNBW-0 showed a higher PA content than Control-0 on days 4 and 10. MNBW-0 was significantly higher than Control-0 by 10.6 % on day 4 ($p = 0.001 < 0.05$), and 6.3 % on day 10 ($p = 0.046 < 0.05$). Similarly, MNBW-1.25 and MNBW-2.5 displayed higher PA levels than the corresponding control groups on days 10, 16, and 22. Specifically, MNBW-1.25 was higher than Control-1.25 by 2.8 % on day 10 ($p = 0.464 > 0.05$), 12.2 % on day 16 ($p = 0.008 < 0.05$), and 2.0 % on day 22 ($p = 0.074 > 0.05$). MNBW-2.5 was higher than Control-2.5 by 13.3 % on day 10 ($p = 0.018 < 0.05$), 4.8 % on day 16 ($p = 0.123 > 0.05$), and 5.3 % on day 22 ($p = 0.372 > 0.05$). These findings suggest that MNBW facilitated alkalinity recovery under HPr inhibition conditions, thereby improving the buffering capacity in AD process.

To comprehensively evaluate the stability of the AD system, we introduced two additional assessment indicators: intermediate alkalinity (IA)/PA and VFA/TA, both of which are widely applied to assess the stability of AD systems (Feng et al., 2020). Fig. 3d shows the overall trend of IA/PA throughout the AD process, which gradually decreased before stabilisation. In the early phase of AD, the IA/PA ratios for Contol-2.5/MNBW-2.5 were both > 2.5, indicating a poor buffering capacity. The IA/PA ratio gradually fell below 1 by day 16, reflecting severe HPr inhibition. Compared to MNBW-2.5, Contol-2.5 was significantly higher on days 10, 16, and 22 ($p < 0.05$). Specifically, Contol-2.5 was significantly higher than MNBW-2.5 by 17.4 % on day 10 ($p = 0.001 < 0.05$), 9.1 % on day 16 ($p = 0.042 < 0.05$), and 7.8 % on day 22 ($p = 0.005 < 0.05$), suggesting that MNBW aided in restoring the stability under severe HPr inhibition. Similar trends were observed for MNBW-1.25 and MNBW-0, which exhibited lower IA/PA ratios than Control-1.25 and Control-0 on days 10, 16, and on days 4, 10, respectively. These results further indicate that MNBW helps reduce the IA/PA ratio under HPr inhibition.

The trends in VFA/TA values for all reactors are shown in Fig. 3e. The initial VFA/TA values for Contol-2.5/MNBW-2.5 and Control-5/MNBW-5 were both > 1.5, indicating poor stability during the early AD phase. The VFA/TA value for Control-1.25 was lower than MNBW-1.25 on day 4, but higher on days 10 and 16, suggesting that MNBW facilitated VFAs production in the early stages and subsequently promoted HPr decomposition and increased alkalinity. This result is consistent with the trends in HPr and VFAs concentrations observed in Fig. 1. The VFA/TA values for MNBW-2.5 were lower than Contol-2.5 on days 4, 16, and 22, further indicating that MNBW was beneficial to improve the buffering capacity in AD process.

In conclusion, these results indicate that MNBW is beneficial for restoring the buffering capacity in AD process. Enhanced system stability in the presence of HPr inhibition may be attributed to HPr degradation by MNBW, thereby preventing VFAs accumulation.

3.4. Soluble organics and TAN

This study focused on the dynamics of soluble carbohydrates and proteins to clarify the effect of MNBW on soluble organics under conditions of HPr inhibition. Fig. 4a shows the behaviour of soluble carbohydrates throughout the AD process under varying degrees of HPr inhibition. The concentrations of soluble carbohydrates in Control-5/MNBW-5 rapidly increased during the early hydrolysis phase of AD, but subsequently showed no significant consumption, indicating that severe HPr inhibition may have affected further degradation of soluble organics, such as carbohydrates. In Contol-2.5/MNBW-2.5, soluble carbohydrates accumulated during the early AD phase and were rapidly consumed later. This behaviour is due to the high concentration of HPr, which initially impeded the consumption of soluble carbohydrates. As HPr inhibition was gradually alleviated, these carbohydrates were utilised. Notably, soluble carbohydrates in MNBW-2.5 were consistently lower than Contol-2.5 on days 28 and 34, suggesting that MNBW supplementation accelerated the degradation of soluble organics, including carbohydrates. Specifically, MNBW-2.5 was lower than Control-2.5 by 7.4 % on day 28 ($p = 0.032 < 0.05$), and 9.6 % on day 34 ($p = 0.122 > 0.05$). When the concentration of HPr was low, the consumption of soluble carbohydrates was not significantly inhibited, and the carbohydrates rapidly decomposed during the early AD phase. These results further confirm that MNBW promotes the degradation and consumption of soluble carbohydrates under HPr inhibition. The trends in soluble protein concentrations across all reactors during AD are shown in Fig. 4b. In Control-5/MNBW-5, soluble proteins were not effectively consumed during AD. Conversely, the other groups showed an initial increase, followed by a decrease in soluble protein levels, which mirrored the changes observed in soluble carbohydrates.

To ascertain whether ammonia inhibition occurred, TAN concentrations were measured across all reactors. The TAN concentrations in all reactors increased gradually in the initial phase before stabilising, with no reactor exceeding 1400 mg/L (Fig. 4c). Preliminary results indicated that ammonia inhibition did not occur during the AD process and that the inhibiting factors in this work were attributed to HPr inhibition. Furthermore, the concentration of free ammonia nitrogen in all reactors remained below 50 mg/L throughout the AD process (Fig. 4d), further confirming the absence of significant ammonia inhibition.

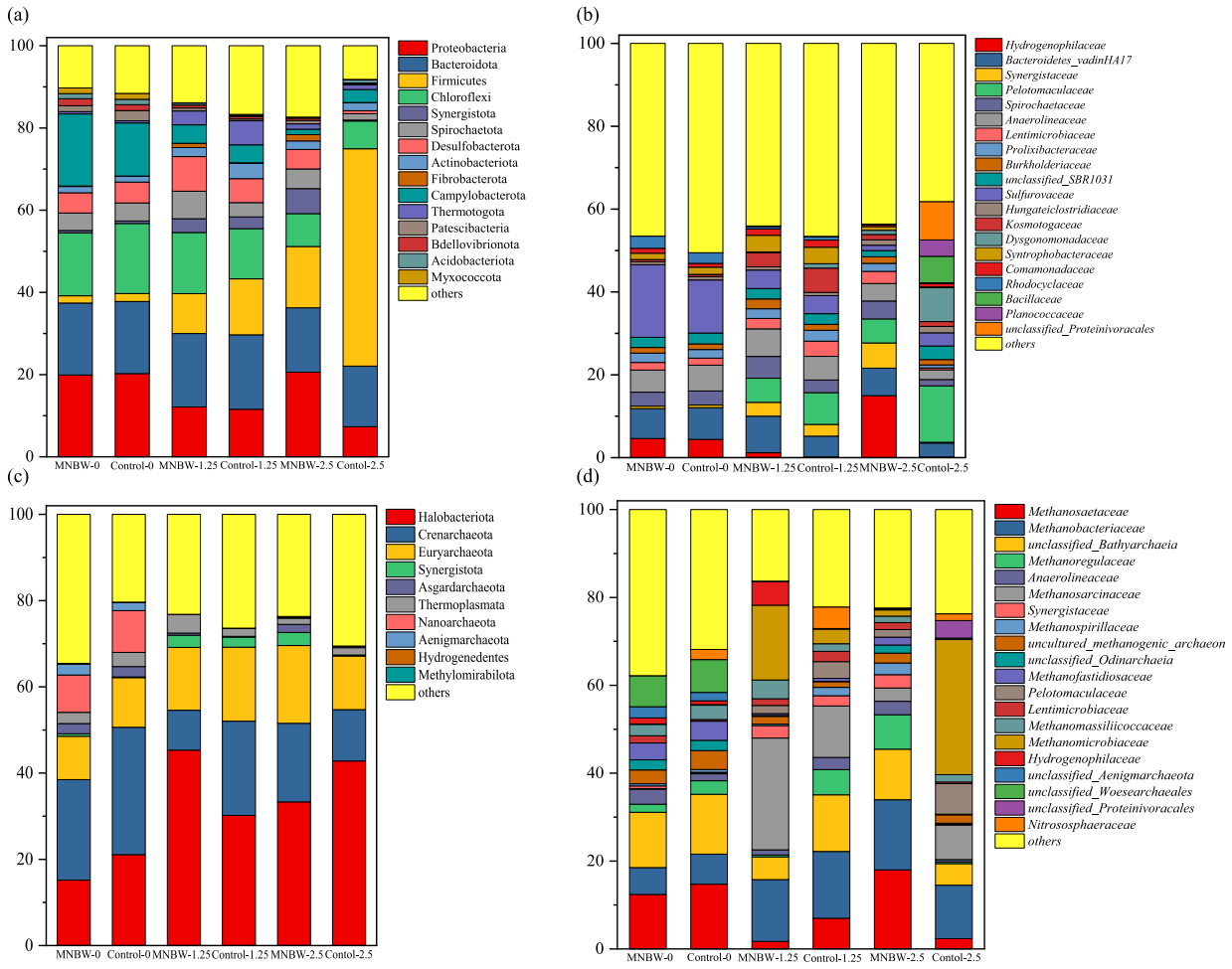


Fig. 6. Microbial community structure. (a) Phylum level of bacteria. (b) Generic level of bacteria. (c) Phylum level of archaea (d) Generic level of archaea.

3.5. Enzyme activity

Hydrolytic enzymes are particularly important in AD system (Pang et al., 2020). Furthermore, during methanogenesis, the activity of certain key enzymes substantially influences methanogen growth and methane yield (Li et al., 2024a). The activities of the four types of extracellular hydrolytic enzymes in MNBW-2.5 were significantly higher than those in the Contol-2.5 (The p values range from 0.002 to 0.023, less than 0.05), with increases ranging from 7.0 % to 13.8 % (Fig. 5a). Additionally, the activities of these extracellular hydrolytic enzymes in MNBW-1.25 and MNBW-0 were also significantly higher than those of Control-1.25 and Control-0, respectively, and the p values range from 0.004 to 0.046, less than 0.05. These findings further validate the conclusion that MNBW enhances the extracellular hydrolytic enzyme activity in the presence of HPr inhibition. Moreover, the experimental results for coenzyme F420 illustrated that F420 activity in MNBW reactors was higher than control reactors (Fig. 5b). Specifically, the F420 activity in MNBW-0 increased by 21.0 % compared to that in Control-0 ($p = 0.001 < 0.05$), which was greater than the increase observed for MNBW-1.25 ($p = 0.019 < 0.05$) and MNBW-2.5 ($p = 0.001 < 0.05$) relative to their respective control group. This suggests that HPr inhibition are unfavourable for MNBW to enhance methanogen activity and methane production. These results reveal that despite HPr inhibition, MNBW could still promote hydrolysis and methanogenesis. Enhanced methanogens are better able to tolerate a certain concentration of VFAs and can convert VFAs, including HPr, into methane and carbon dioxide more quickly, thereby reducing the accumulation of HPr and easing the inhibition of VFAs (Li et al., 2021a).

3.6. Microbial community structure

To find out the reasons by which MNBW accelerates HPr degradation, alleviates HPr inhibition, and increases methane production, we compared the microbial community structure differences between the reactors with MNBW and control groups. Since the AD process of Control-5/MNBW-5 failed, the other six groups of samples were mainly concerned, namely Control-2.5/MNBW-2.5, Control-1.25/MNBW-1.25 and Control-0/MNBW-0. In the control groups, the abundance of archaea and bacteria showed a downward trend with an increase in HPr, indicating the negative effect of HPr inhibition in $R_{Control}$ (see supplementary material). Different from the control group, the bacterial abundance of MNBW-1.25 was higher than that of MNBW-0, indicating that MNBW can play a positive role in improving bacterial abundance when HPr inhibition is not serious. The bacterial abundance of MNBW-1.25, however, was lower than that of MNBW-2.5, suggesting that MNBW could not improve bacterial abundance when HPr inhibition was severe.

The dominant bacteria were *Proteobacteria*, *Bacteroidetes*, and *Firmicutes* at the phylum level of bacteria (Fig. 6a). Reportedly, these microorganisms fermented with polysaccharide substances in the system and participated in biodegradation of macromolecular organics to provide a small molecular carbon source for methanogens (Strąpoć et al., 2011; Wu et al., 2024). *Hydrogenophilaceae* and *Bacteroidetes_vadinHA17* are participated in the denitrification and degradation of various complex organics (Sun et al., 2015). *Bacteroidetes_vadinHA17* used more polysaccharides and sterols in the system for acid production by fermentation and accelerates the reaction process (Baker et al., 2020).

The relative abundance (RA) of *Bacteroidetes_vadinHA17* in MNBW-1.25 was 8.8 %, which was higher than Control-1.25 by 76.2 %. Similarly, the RA of MNBW-2.5 (6.7 %) was higher than Control-2.5 by 97.0 %. These results suggest that MNBW promotes the consumption of proteoglycans and accelerates methane production. As for *Hydrogenophilaceae*, the RA of MNBW-0, MNBW-1.25 and MNBW-2.5 were 4.6 %, 1.2 % and 14.9 %, respectively, higher than those of the corresponding controls. Previous studies have shown that *Sulfurovaceae* and methanogens complement nutrient metabolism (Nie et al., 2021). *Sulfurovaceae* gradually decreased with increasing HPr in R_{MNBW} and $R_{Control}$ (Fig. 6b), indicating that high HPr concentrations were not conducive to the nutritional metabolic pathways of methanogens, thereby inhibiting methane production. In addition, *Sulfurovaceae* in MNBW-1.25 and MNBW-0 were higher than Control-1.25 and Control-0, respectively. Specifically, MNBW-0 (17.5 %) was higher than Control-0 by 36.7 %. Similarly, MNBW-1.25 was higher than Control-1.25 by 10.8 %. These results indicates that MNBW supplementation is of positive significance for optimising the nutritional metabolic pathways of methanogens and increasing methane production when HPr inhibition is not severe.

In conclusion, *Hydrogenophilaceae* and *Bacteroidetes_vadinHA17* were the dominant bacteria and contributed greatly to accelerating the consumption of proteoglycans and promoting acidogenic methanogenesis. A higher concentration of HPr inhibited methane production, whereas MNBW relieved HPr inhibition, promoted proteoglycan consumption, and accelerated methane production.

The main archaeal phyla, including *Halobacterota*, *Euryarchaeota*, and *Crenarchaeota* were all methanogenic archaea (Yutin et al., 2008). *Halobacterota* had the largest proportion (18.3 %–44.5 %). *Crenarchaeota* was the second most dominant species (6.4 %–22.1 %), whereas the proportion of *Euryarchaeota* was approximately 8.5 %–15.4 % (Fig. 6c). As shown in Fig. 6d, the dominant species were *Methanosaetaceae*, *Methanobacteriaceae*, unclassified_*Bathyarchaeia*, and *Methanosaricinaceae*. *Methanosaetaceae* was the most common species that produce methane from acetate (Smith and Ingram-Smith, 2007). It decreased gradually with an increase in HPr in $R_{Control}$, indicating that high-concentration HPr inhibited *Methanosaetaceae* and caused systematic acidification, affecting the methane production process. The RA of *Methanosaetaceae* in MNBW-2.5 was 18.0 %, which was higher than that in Control-2.5 (2.3 %), suggesting that supplementing MNBW enhanced acetate consumption, thus alleviating inhibition and increasing methane production.

In conclusion, *Halobacterota*, *Euryarchaeota*, and *Crenarchaeota* are the three dominant methanogens. A high HPr concentration caused systematic acidification and affected the methanogenic process. Supplementation with MNBW alleviates HPr inhibition to a certain extent. However, excessive HPr inhibited the abundance of acetic acid-methanogenic microorganisms.

3.7. Possible underlying mechanisms related to MNBW alleviating HPr inhibition

To clarify the underlying mechanisms by which MNBW alleviates HPr inhibition, the composition and characteristics of Air-MNBW and its role in accelerating HPr degradation and ultimately increasing methane production were analyzed in this work. As we all know, Air-MNBW is composed of water and air-MNBs, which mainly contain O_2 and N_2. Previous studies have shown that both O_2-MNBW and N_2-MNBW can promote AD process and increase methane production (Wang et al., 2020, Yang et al., 2019). Although the introduction of O_2 may contradict the anaerobic environment of AD, a number of reports have shown that O_2-MNBW is conducive to the degradation of refractory organics and increase the activity of facultative bacteria, thus promoting the AD process and increasing methane production (Wang et al., 2021; Wang et al., 2023). In addition, studies have shown that the micro-oxygen environment is conducive to the AD process (Nguyen and Khanal, 2018; Nguyen et al., 2019). Therefore, the introduction of O_2 in Air-MNBs in this study may play a positive role in the AD process. As for MNBs, it has the characteristics of large specific surface area, strong stability and rich in free radicals. Its rupture releases physical energy (such as shear force) and produces a large number of reactive oxygen species, which is conducive to the degradation of refractory organics (Jia et al., 2023; Wang et al., 2019).

The composition of Air-MNBW determines that it has characteristics that DW does not have. It has higher spin–spin relaxation times, zeta potential and electron transport system (ETS) activity, etc, which makes MNBW have high mobility, strong solubility and high mass transfer efficiency (Chuenchart et al., 2021; Liu et al., 2019). The introduction of MNBs, O_2 and water molecules with high mobility during AD could affect HPr inhibition. Firstly, in the hydrolysis stage of AD process, Air-MNBW improved the activity of hydrolase and promoted the degradation of soluble carbohydrates/proteins (Fig. 4 and Fig. 5). Subsequently, in the acidogenesis stage, the supplementation of Air-MNBW accelerated VFAs production (Fig. 1). After that, in the acetogenesis stage, Air-MNBW promoted the degradation of HPr to acetate (Fig. 1). Finally, in the methanogenesis stage, the supplementation of Air-MNBW increased the activity of key enzymes (such as F420) and methanogens (Fig. 5). Moreover, a previous study has shown that MNBW supplementation enhances the activity of hydrogenotrophic methanogens (HM) and acetoclastic methanogens (AM) (Wang et al., 2020). The increased activity of HM and AM could tolerate the toxicity caused by a certain concentration of VFAs, accelerate the degradation of VFAs (including HPr), and alleviate the inhibition of HPr (Li et al., 2021a). In addition, previous studies have shown that the supplementation of MNBW can improve ETS activity and mass transfer efficiency (Chuenchart et al., 2021; Temesgen et al., 2017). This result suggests that Air-MNBW may strengthen the DIET between methanogens and syntrophic bacteria, and is conducive to the mitigation of HPr inhibition. Further studies are needed to elucidate the underlying mechanisms by which MNBW alleviates HPr inhibition.

4. Conclusion

The role of MNBW in accelerating HPr degradation, enhancing methane production, and optimizing methanogenic pathways was systematically studied in this work. Results suggested that MNBW accelerated HPr degradation and improved system stability. In addition, cumulative methane production increased by 13.5 % to 18.4 % when MNBW was introduced compared with control group. Moreover, supplementing MNBW boosted the activity of related enzymes. MNBW also optimised the microbial community structure and enhanced methanogen tolerance to HPr inhibition. MNBW, being a pollution-free, chemical-free, and low-cost technology, does not require complex equipment or high energy consumption. As an environmentally friendly approach, MNBW offers a novel solution for addressing HPr-related inhibition, laying the foundation for broader industrial applications in waste treatment and bioenergy production, with significant potential for further development.

CRediT authorship contribution statement

Yujie Fan: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yuqi Liu:** Validation, Investigation, Formal analysis, Data curation. **Ziyang Zhang:** Resources, Investigation, Data curation. **Jinke Sun:** Validation, Formal analysis. **Xiaojing Yang:** Writing – review & editing, Conceptualization. **Peng Deng:** Validation, Formal analysis. **Haibo Yang:** Resources, Formal analysis. **Ziwen Zhao:** Writing – review & editing, Validation, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2025.132425.

Data availability

Data will be made available on request.

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